

Lucas A. Saavedra

SOFTWARE ENGINEER · AI RESEARCHER

Buenos Aires, Argentina

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Summary

I am a software engineer specializing in back-end development, artificial intelligence, computational neuroscience, and machine learning. My experience includes deep learning for neurobiological data analysis and back-end development for cybersecurity applications. Through my skills, I have published in international scientific journals and created open-source software, demonstrating proficiency and innovation in both engineering and science.

Skills

Soft Skills

Effective communication, Teamwork, Problem-solving, Critical thinking, Resilience in complex challenges, Stress management and frustration, Fast adaptability

DevOps

AWS, Docker, CircleCI

Technologies

Django, REST API, TensorFlow, PyTorch, Pandas, NumPy, SQL, MongoDB, Redis

Programming

Python, JAVA, C, C++, R, LaTeX, MATLAB

Education

Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

SOFTWARE ENGINEERING

2018 - 2023

- **Final work:** Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories. (Grade: 10)

Experience

Deloitte

Buenos Aires, Argentina

BACK-END DEVELOPER

Jan. 2021 - Jun. 2022

- Developed Codex Gigas, a malware "DNA" search engine that identifies patterns and characteristics for malware hunting using REST APIs with Python.
- I worked in Agile environments.
- Built ImageSimilarity, a tool to classify images based on similarity using the metric SSIM.
- Developed applications for ThreatConnect using Python, Docker, MongoDB, and SQL.
- **Technologies used: Python, Docker, MongoDB, SQL, Redis.**

BIOMED UCA-CONICET, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

RESEARCH ASSISTANT

Jan. 2022 - PRESENT

- Implemented Deep Learning models for static and dynamic analysis of proteins in cells.
- Analyzed spatial distribution and density of proteins using super-resolution microscopy (STED, STORM, etc) and MATLAB.
- Developed an efficient and autonomous GNN-based algorithm in TensorFlow to quantify dynamic particle clustering in molecules data.
- **Technologies used: Python, Java, Tensorflow, Pandas, PyTorch, Docker, NumPy, MongoDB, MATLAB.**

Facultad de Ingenieria, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

ASSISTANT PROFESSOR

Mar. 2023 - PRESENT

- "Introduction to Data Science": Artificial Intelligence concepts, Neural Networks, Scientific Method.
- "Data-Oriented Programming": Object-Oriented Programming, Python, Java, NumPy, Pandas, UML Software Design.
- "Discrete Mathematics": Number Theory, Graph Theory, Proof Methods, Boolean Algebra.

UpWork

FREELANCER

Dec. 2023 - PRESENT

- Fixed and hourly contracts.
- Web scraping with Python and Selenium in AWS EC2.
- YOLO training to detect basketball players.
- **Technologies used: Python, YOLO, Pandas, Docker, SQL, EC2 AWS, Selenium.**

Certifications and Courses

- 2020 **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
- 2020 **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera
- 2021 **Machine Learning**, Stanford Online, Coursera
- 2021 **Programming Foundations: Design Patterns (2013)**, Linkedin Learning
- 2021 **R Programming**, Johns Hopkins University, Coursera
- 2022 **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera
- 2022 **DeepLearning.AI Tensorflow Developer**, DeepLearning.AI, Coursera
- 2024 **Deep Neural Networks with PyTorch**, IBM, Coursera
- 2025 **Introduction to Statistics**, Stanford Online, Coursera

Publications and conferences

- 2022 **Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies**, International Journal of Molecular Sciences
- 2023 **The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor**, 11th International Weber Symposium *Punta del Este, Uruguay*
- 2024 **A supervised graph-based deep learning algorithm to detect and quantify clustered particles**, Nanoscale
- 2024 **Machine learning empowers static and dynamics clustering characterization of transmembrane receptors**, Building Bridges in Computational Biophysics (BBCB) v3.0